

5. Show that uniform sampling of a signal leads to a periodic Fourier transform.
6. What is the period of the Fourier transform of a sampled signal?
7. Show that a periodic signal has a sampled Fourier transform.
8. What is the sampling period of the Fourier transform of a periodic signal?

$$s) \quad p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

$$X_s(t) = x(t) \cdot p(t) = x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT) = \sum_{n=-\infty}^{\infty} x(nT) \delta(t - nT)$$

$$X_s(\omega) = \int_{-\infty}^{\infty} x_s(t) e^{-i\omega t} dt = \int_{-\infty}^{\infty} \left(\sum_{n=-\infty}^{\infty} x(nT) \delta(t - nT) \right) e^{-i\omega t} dt =$$

$$= \sum_{n=-\infty}^{\infty} x(nT) \int_{-\infty}^{\infty} \delta(t - nT) e^{-i\omega t} dt =$$

$$= \sum_{n=-\infty}^{\infty} x(nT) e^{-i\omega nT}$$

$$\omega_s = \frac{2\pi}{T}$$

$$X_s(\omega + \omega_s) = \sum_{n=-\infty}^{\infty} x(nT) e^{-i(\omega + \omega_s)nT} = \sum_{n=-\infty}^{\infty} x(nT) e^{-i\omega nT} e^{-i\omega_s nT} =$$

$$-i\omega_s nT = -i \frac{2\pi}{T} nT = -i2\pi n$$

$$e^{-i2\pi n} = \cos(2\pi n) - i \sin(2\pi n) = 1 - i \cdot 0 = 1$$

$$= \sum_{n=-\infty}^{\infty} x(nT) e^{-i\omega nT}$$

6) the period of the Fourier transform is f_s

7) $x(t)$ with $T_0 = \frac{2\pi}{\omega_0}$

$$\text{Fourier expansion: } x(t) = \sum_{k=-\infty}^{\infty} c_k e^{i\omega_0 k t}$$

$$c_k = \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t) e^{-ik\omega_0 t} dt$$

$$\begin{aligned}
 X(\omega) &= \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt = \int_{-\infty}^{\infty} \left(\sum_{k=-\infty}^{\infty} c_k e^{ik\omega_0 t} \right) e^{-i\omega t} dt = \\
 &= \sum_{k=-\infty}^{\infty} c_k \int_{-\infty}^{\infty} e^{-i(\omega - k\omega_0)t} dt = \sum_{k=-\infty}^{\infty} c_k 2\pi \delta(\omega - k\omega_0) \\
 &= 2\pi \sum_{k=-\infty}^{\infty} c_k \delta(\omega - k\omega_0) \quad \text{periodic}
 \end{aligned}$$

8) the sampling period = f_0 where $f_0 = \frac{1}{T_0}$ of the original function

10. What is the frequency resolution of the discrete Fourier transform of a T sampled signal?

11. What are the frequency limits (min/max) of the discrete Fourier transform of a T sampled signal?

10) freq. resolution $\Delta f = \frac{1}{NT} = \frac{f_s}{N}$

$\nwarrow$ $\nearrow$
 nr. of samples sampling period

11) $f_{\max} = \frac{f_s}{2}$
 $f_{\min} = -\frac{f_s}{2}$ OR 0 depends on what context